3.5 Measures to reduce the vulnerability

Measures to reduce the vulnerability of the elements at risk consist of "passive" solutions which are not intended to prevent the triggering of the landslide but to reduce the resulting degree of loss. They can be subdivided in two main categories, depending on the approach followed to achieve this objective:

- Measures to increase the resistance of elements at risk (reduction of vulnerability s.s.) – existing structures can be strengthened; for new structures, the potential effects of impact from landslide material can be taken into account from the outset. This approach is typically applicable only in relation to relatively shallow slides, since it is practically impossible to build structures capable of withstanding the impact form larger landslides (Figure 14 and Figure 15).
- Measures to stop or to deviate the path of the landslide debris (reduction of vulnerability s.l.) - Works can be carried out to intercept and block or at least to deviate or to slow down the sliding materials. This type of works relates mainly to the fall of massive blocks or to flows of all types, in those cases where a large slope is affected and stabilization is not feasible for environmental impact reasons or because of cost.

Measures to reduce the vulnerability through the increase of structural resistance may be summarized as follows:

- Strengthening of shallow foundations and improved structural design to withstand predicted permanent ground displacements;
- Deep foundations properly designed to accommodate the landslide effect;
- Deep anchoring of foundation elements;
- Combination of the above three approaches.

Another important parameter affecting the strengthening approach is the fact that any approach is strongly case-depended, in the sense that the characteristics of the structure, its relative position within the landslide zone and the regolith-rock properties play an important role in any strengthening decision. This is why the relevant research and design practice is rather fragmented. Consequently it is practically impossible to define, in a general way, the improvement in the vulnerability (in quantitative terms) obtained by increasing the resistance of different parts of the structure, because de facto this is a case depended evaluation.

Measures to stop or to deviate the path of the landslide debris resistance relate to the following landslide types (Figure in Appendix 1.1):

a) Earth or debris flows of any type;
b) Toppling, rumbling or free falling rocks of various sizes.

They should be foreseen when the general stabilization of the landslide is not feasible from technical, environmental and financial point of view.